

2.3 Limits Laws

Limits Laws Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exists. Then

$$1. \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$2. \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$3. \lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$$

$$4. \lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

5. If $\lim_{x \rightarrow a} g(x) \neq 0$ then

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

$$6. \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$$

$$7. \lim_{x \rightarrow a} c = c \quad ; \quad 8. \lim_{x \rightarrow a} x = a$$

$$9. \lim_{x \rightarrow a} x^n = a^n, \quad n \in \mathbb{N}$$

$$10. \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \quad (\text{If } n \text{ is even, } a > 0)$$

$$11. \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}, \quad n \in \mathbb{N}$$

(If n is even then $\lim_{x \rightarrow a} f(x) > 0$)

Direct substitution: If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

If $f(x) = g(x)$ when $x \neq a$ then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$ provided the limit exist.

Example 1:

$$(a) \lim_{x \rightarrow 3} (3x^2 + 2\cos(\frac{\pi}{x}) + 5)$$

$$(b) \lim_{x \rightarrow 2} \frac{x-2}{3 - \sqrt{7+x}}$$

Example 1:

$$(a) \lim_{x \rightarrow 3} (3x^2 + 2\cos(\frac{\pi}{x}) + 5) = 3 \times 3^2 + 2 \times \cos(\frac{\pi}{3}) + 5$$

Since $\lim_{x \rightarrow 3} x^2$, $\lim_{x \rightarrow 3} \cos(\frac{\pi}{x})$ and $\lim_{x \rightarrow 3} 5$

all exist.

$$(b) \lim_{x \rightarrow 2} \frac{x-2}{3-\sqrt{7+x}} \quad \left(\text{form } \frac{0}{0} \right)$$

We need to rationalize the denominator, we multiply the numerator & denominator of this rational by the conjugate of " $3-\sqrt{7+x}$ ", that is " $3+\sqrt{7+x}$ ".

("a-b" and "a+b" are conjugate.

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x-2}{3-\sqrt{7+x}} &= \lim_{x \rightarrow 2} \left(\frac{x-2}{3-\sqrt{7+x}} \times \frac{3+\sqrt{7+x}}{3+\sqrt{7+x}} \right) \\ &= \lim_{x \rightarrow 2} \frac{(x-2)(3+\sqrt{7+x})}{(3-\sqrt{7+x})(3+\sqrt{7+x})} \end{aligned}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(3+\sqrt{7+x})}{9 - (\sqrt{7+x})^2}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(3+\sqrt{7+x})}{9 - (7+x)}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(3+\sqrt{7+x})}{2-x}$$

$$= \lim_{x \rightarrow 2} -(3+\sqrt{7+x})$$

$$= -(3+\sqrt{7+2}) = -6$$

note that $\frac{3+\sqrt{7+x}}{3+\sqrt{7+x}} = 1$

so $f(x) \times 1 = f(x)$

The limit is indeed the one we are looking for.

"To rationalize" the numerator or denominator is sometimes necessary

$$f(x) = \frac{\sqrt{x+1} - 1}{x} \quad \left(\text{form } \frac{0}{0} \right)$$

$$\begin{aligned} f(x) &= \frac{\sqrt{x+1} - 1}{x} \times \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} \\ &= \frac{(\sqrt{x+1})^2 - 1^2}{x\sqrt{x+1} + x} = \frac{x+1-1}{x(\sqrt{x+1}+1)} = \frac{x}{x(\sqrt{x+1}+1)} \end{aligned}$$

$$f(x) = \frac{1}{\sqrt{x+1} + 1}$$

Hence

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+1} + 1} = \frac{1}{2}$$

Theorem : If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a , then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

The squeeze theorem : If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

Example 2 : Find

(a) $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right)$

(b) $\lim_{x \rightarrow 1} \frac{\sin(\sqrt{x} - 1)}{x - 1}$

Example 2:

(a) Observe that $\lim_{x \rightarrow 0} \cos\left(\frac{1}{x}\right)$ does not exist, but

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1 \quad \forall x \neq 0 \quad (1)$$

for $x > 0$ $(1) \Rightarrow -x \leq x \cos\left(\frac{1}{x}\right) \leq x \quad (2)$

by squeeze theorem with $f(x) = -x$ and $h(x) = x$, since $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} g(x) = 0$

$$\text{then } \lim_{x \rightarrow 0^+} x \cos\left(\frac{1}{x}\right) = 0$$

for $x < 0$ $(1) \Rightarrow -x \geq x \cos\left(\frac{1}{x}\right) \geq x \quad (3)$

with $f(x) = x$ and $h(x) = -x$, since

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} h(x) = 0, \text{ by squeeze theorem}$$

$$\lim_{x \rightarrow 0^-} x \cos\left(\frac{1}{x}\right) = 0$$

$$\text{Hence } \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$

(b) $\lim_{x \rightarrow 1} \sinh(\sqrt{x}-1) = 0$ and $\lim_{x \rightarrow 1} (x-1) = 0$
so we have the indeterminate form $\left(\frac{0}{0}\right)$

~~$\forall x \geq 0$~~ $\forall x \geq 0 \quad -1 \leq \sinh(\sqrt{x}-1) \leq 1 \quad (1)$

$\forall x \in [0, 1) \quad , \quad \frac{1}{x-1} < 0 \quad \text{so}$

$$(1) \Leftrightarrow -\frac{1}{x-1} \geq \frac{\sinh(\sqrt{x}-1)}{x-1} \geq \frac{1}{x-1}$$

Since $\lim_{x \rightarrow 1^-} -\frac{1}{x-1} = +\infty$ (does not exist)

and $\lim_{x \rightarrow 1^-} \frac{1}{x-1} = -\infty$ (does not exist)

The squeeze theorem cannot be used!!

We will deal with this limit
at a later stage

$$\left(\frac{\sinh(\sqrt{x}-1)}{x-1} = \frac{\sinh(\sqrt{x}-1)}{\sqrt{x}-1} \times \frac{1}{\sqrt{x}+1} \right)$$

Example 3 : If $2x^2 \leq g(x) \leq 3x+2$

for all $x \in \mathbb{R}$, determine $\lim_{x \rightarrow 2} g(x)$.

$\therefore$ Note that $\lim_{x \rightarrow 2} 2x^2 = 8$ and

$\lim_{x \rightarrow 2} (3x+2) = 3 \times 2 + 2 = 8$. By the

Squeeze theorem, $\lim_{x \rightarrow 2} g(x) = 8$.